exploration rate. The spectral properties of the operator $\mathcal{L}_{\mathrm{LD}}^{\dagger}$ guarantee that we will explore the entire phase space (ergodicity), while the relatively small perturbations to the operator induced by numerical discretization are hoped not to significantly alter the rate of search. Ultimately, pushing the timestep up is the only way to breach timescale gaps when computing trajectories, although this comes at the cost of corruption of the long-time averages.

The self-diffusion coefficient gives a metric quantifying the diffusion rate. It is often used as a way to compare the rate of phase space exploration between methods, and is typically calculated using the integral of the velocity auto-correlation function. However, in theory one can construct arbitrary methods to artificially scale the velocity auto-correlation function, hence giving inaccurate diffusion constants: the momentum sampled may not be the momentum we use to propagate the position with.

Indeed, calculating the temperature of the system from an average of kinetic energy by

$$\mathrm{N_d k_B T} = \left\langle \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p} \right\rangle, \tag{7.41}$$

gives a similar problem. Using Proposition 6.1 we may formulate many alternative functions whose averages are proportional to the system temperature [193, 211, 319]. Such observables are normally based on the periodic forces of the system in order to work in periodic boundary conditions; in the following experiments we shall use the configurational temperature

$$\mathrm{N_d k_B T} = \langle \boldsymbol{q} \cdot \nabla U(\boldsymbol{q}) \rangle. \tag{7.42}$$

Were one able to solve the dynamics exactly, the kinetic and configurational temperatures would of course be equal. However, using a numerical method in the large-timestep regime we instead sample expectations with respect to the perturbed density $\hat{\rho}$, that may introduce discrepancies between configurational and kinetic temperatures. Often configuration-based temperature calculation is more useful and relevant for assessing the quality of configurational sampling methods.

In a similar way, the speed of exploration of the space should not be determined solely from functions of momentum. Far more reliable is to compute configurational quantities such as barrier crossing rates, passage times to reach some target region of phase space or the diffusion constant through an Einstein relation (which uses the position data).

The friction parameter is free in Langevin dynamics when it is used purely as a sampling method; we know that for any $\gamma > 0$ we will obtain states from an approximate canonical distribution, though it is unclear what value of γ we should choose in our simulation. From our analysis in Sect. 7.9.3, we have seen that the ⟦BAOAB⟧ scheme has a superconvergence property in the limit of large friction, giving us higher accuracy $\mathcal{O}(h^4)$ configurational averages. However, it is unclear what the scheme's behavior is in the middle-to-low friction limit, as we did not give